

Fraud Detection and AML Alert Triage

Client: STADIOEquities

Recommendations Report

CAP182: Capstone Project SS2

Introduction

This report presents recommendations to STADIOEquities based on the modelling work carried out on a publicly available proxy dataset. Two models were built and evaluated for the task of triaging transaction-monitoring alerts: a Logistic Regression model (Model 1) and a Random Forest model (Model 2). The purpose of the work was to determine whether a machine learning model can reliably separate genuine cases from false positives, thereby reducing the false positive burden that currently strains the compliance department. The recommendations below address which model is preferable, how it can be improved, how it should be adapted for STADIOEquities own data and how the findings compare with the published literature.

1 Which model is best?

Model 2, the Random Forest, is the recommended model. Although Model 1 achieved a higher recall (0.92 against 0.84) which means it caught marginally more fraud at an unacceptable cost. Model 1 flagged 1,533 legitimate transactions as fraudulent, whereas Model 2 flagged only 27, a reduction of more than 98 percent. This is decisive because the false-positive volume is precisely the problem the project sets to solve. Model 2 also outperformed Model 1 on every balanced measure, including precision (0.75 against 0.06), F1 score (0.79 against 0.10) and PR-AUC (0.86 against 0.72). The insight here is that recall alone is a misleading basis for selection in this context: a model that catches almost all fraud is of little practical use if it simultaneously overwhelms investigators with false alarms. Model 2 delivers a workable balance, catching the large majority of fraud while keeping false positives at a level a human team can realistically review.

2 How can the selected model be improved?

There are three improvements recommended with each tied directly to the observed results. First, hyperparameter tuning should be undertaken. The Random Forest was trained with sensible default settings (100 trees and a maximum depth of 20); a systematic search over these values, using cross validation is likely to improve performance further and would place configuration on a firmer evidence base. Second, the decision threshold should be tuned. The current results use the default probability threshold of 0.5, but due to the costs of false positive

and false negative differ for STADIOEquities, adjusting the threshold would allow the business to deliberately choose its preferred point on the precision-recall trade-off rather than accepting the default. Third, the small loss in recall relative to Model 1 (eight additional missed fraud cases) suggests value in exploring cost-sensitive learning or gradient-boosted methods such as XGBoost, which often recover recall without sacrificing precision. Together these steps target the exact weakness the results revealed, which is the recall gap, rather than pursuing generic improvement.

3 Adapting the model for STADIOEquities own data (SS1)

The model was developed on a proxy dataset of anonymised credit-card transactions, because STADIOEquities own data was not available for the study. To deploy the approach on the data requested in SS1, several adaptations are required. The SS1 data request specified compliance alerts with their investigator dispositions and case outcomes; these dispositions (genuine case versus false positive) would become the target label, replacing the proxy dataset's fraud label. The proxy dataset's features are anonymised PCA components, whereas STADIOEquities data contains interpretable features such as transaction type, amount, channel, account tenure, KYC status and device or login signals. The feature engineering would therefore be stronger and more domain driven than was possible on the proxy data and would likely improve performance. The core methodology, however transfers directly: the same stratified train-test split, the same use of SMOTE on the training set only, and the same imbalance-aware evaluation metrics would all apply, because STADIOEquities alert data shares the same rare-event, highly imbalanced structure as the proxy dataset. One additional consideration is that real alert labels may be noisier and subject to delay, which the deployment pipeline would need to accommodate.

4 Alignment with the related literature

The results align closely with the literature reviewed. (Albalawi & Dardouri, 2025) found the random forest to be the strongest of the traditional classifiers for imbalanced credit-card fraud detection and confirmed the value of SMOTE for handling class imbalance; both findings are reproduced here, with the Random Forest clearly outperforming Logistic Regression. (Naser Eddin, et al., 2022) framed alert triage as a model that suppresses false positives while prioritising true positives and reported a large reduction in false positives; the 98 percent reduction achieved in this study is consistent with that outcome. (Gerlings & Constantiou, 2023) emphasised that a triage model must be explainable to investigators, which supports the decision to retain the interpretable Logistic Regression as a baseline and to favour a model whose feature importance can be examined. Some differences should be acknowledged (Naser Eddin, et al., 2022) worked with real, proprietary banking data and used graph-based features, neither of which was available in this study, so their absolute performance is not directly

comparable to the results reported here. This study also used a proxy dataset rather than a genuine AML alert data, meaning the alignment is one of pattern and direction rather than identical figures.

Conclusion

The Random Forest is recommended as the more suitable model for STADIOEquities alert-triage problem, because it dramatically reduces false positives while retaining strong fraud detection. Its performance can be improved through hyperparameter and threshold tuning and by exploring cost-sensitive methods. The approach transfers to STADIOEquities own data with richer, interpretable features but the underlying methodology, and the findings are consistent with the published literature on machine learning for fraud detection and alert triage.

References

- Albalawi, T. & Dardouri, S., 2025. Enhancing credit card fraud detection using traditional and deep learning models with class imbalanced mitigation. *Frontier in Artificial Intelligence*, Volume 8.
- Gerlings, J. & Constantiou, I., 2023. *Machine Learning in Transaction Monitoring: The Prospect of xAI*. Honolulu, Hawaii International Conference on System Science, pp. 3474-3483.
- Naser Eddin, A. et al., 2022. Anti-Money Laundering Alert Optimisation Using Machine Learning with Graphs. *arXiv*.